$\boxed{\text{Feb 3 2026}}$

Ex. $\dot{x} = \underbrace{ax^2}_{f(x)}$

$$\left.\frac{\partial f}{\partial x}\right|_{x=0} = 2ax \Big|_{x=0} = 0$$

Ex. $\dot{x} = ax^3$

$\dot{y} = -y$

Ex. $\dot{x}_1 = x_1 x_2 \qquad = f_1(x_1, x_2)$

$\dot{x}_2 = -x_2 - x_1^2 \quad = f_2(x_1, x_2)$

$$\frac{\partial f}{\partial x} = \begin{bmatrix} x_2 & x_1 \\ -2x_1 & -1 \end{bmatrix}$$

$$\left.\frac{\partial f}{\partial x}\right|_{x=0} = \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix} \qquad \lambda = 0, -1$$

Center Manifold Theorem

When the Jacobian has at least one eigenvalues on the $j\omega$ axis

2D $\lambda_1 = 0$, $\lambda_2 < 0$

$\dot{x}_1 = f_1(x_1, x_2) \qquad \left.\frac{\partial f}{\partial x}\right|_{x=0} = A$

$\dot{x}_2 = f_2(x_1, x_2)$

$\dot{x} = Ax + (f(x) - Ax) \qquad$ Suppose $A = \begin{bmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{bmatrix}$

$$x = \begin{bmatrix} y \\ z \end{bmatrix}$$

$$\overbrace{\qquad\qquad}^{g_1(y,z)}$$

$$\dot{y} = \lambda_1 y + f_1(y,z) - \lambda_1 y$$

$$\dot{z} = \lambda_2 z + \underbrace{f_2(y,z) - \lambda_2 z}_{g_2(y,z)}$$

$$\dot{y} = \lambda_1 y + g_1(y,z)$$

$$\dot{z} = \lambda_2 z + g_2(y,z)$$

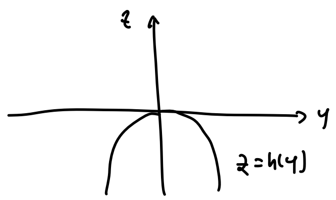
$$g_i(0,0) = 0 \qquad i = 1,2$$

$$\frac{\partial g_i}{\partial y}(0,0) = 0 \qquad i = 1,2$$

$$\frac{\partial g_i}{\partial z}(0,0) = 0$$

<u>Thm</u> There exists a invariant "manifold" $z = h(y)$ defined in a neighborhood of the origin such that

$$h(0) = 0, \ \frac{\partial h}{\partial y}(0) = 0$$



<u>Ex.</u>

$$\dot{x}_1 = x_1 x_2 \qquad = f_1(x_1, x_2)$$

$$\dot{x}_2 = -x_2 - x_1^2 \qquad = f_2(x_1, x_2)$$

$$A = \begin{bmatrix} \overset{\lambda_1}{0} & 0 \\ 0 & \underset{\lambda_2}{-1} \end{bmatrix} \qquad \begin{matrix} y = x_1 \\ z = x_2 \end{matrix}$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = I \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$g_1(y,z) = f_1(y,z) - A_1 y$$

$$= yz - (0)y$$

$$= yz$$

$$g_2(y,z) = f_2(y,z) - A_2 z$$

$$= -z - y^2 + z$$

$= -y^2$

$\dot{y} = 0 + \underbrace{yz}_{g_1}$

$\dot{z} = -z \underbrace{- y^2}_{g_2}$

Thm $\Rightarrow$ $\exists$ invariant manifold $z = h(y)$ with $h(0) = 0$, $\frac{\partial h}{\partial y}(0) = 0$

Thm2: If $y = 0$ is asymptotically stable (resp. unstable) for the "reduced" system then $x = 0$ is asymptotically stable (resp. unstable) for the full system $\dot{x} = f(x)$

Q1: How to find h?

Q2: "reduced" system

Q3: stability of "reduced" system (nonlinear)

Q1: "invariant" manifold $\Rightarrow h$

$w := z - h(y)$

$z(0) = h(y(0)) \Rightarrow z(t) = h(y(t)) \quad \forall t$

$w(0) = 0 \quad \Rightarrow w(t) = 0 \quad \forall t$

$w(0) = 0 \quad \Rightarrow \dot{w} = 0 \quad \forall t$

$\dot{w} = \dot{z} - \frac{\partial h}{\partial y}(y)\, \dot{y}$

$= \lambda_2 z + g_2(y,z) - \frac{\partial h}{\partial y}(\overset{0}{\cancel{\lambda_1}} y + g_1(y,z)) = 0 \quad$ for $z = h(y)$

$\lambda_2 h(y) + g_2(y, h(y)) - \frac{\partial h}{\partial y} g_1(y, h(y)) = 0$

In our example, $\lambda_2 = -1$

$g_1 = y\, h(y)$

$g_2 = y^2$

$$\rightarrow -h(y) + y^2 - \frac{\partial h}{\partial y}\, y\, h(y) = 0$$

Try: $h(y) = ay^2 + O(y^3)$

$$\frac{\partial h}{\partial y} = 2ay + O(y^2)$$

$$-ay^2 - O(y^3) + y^2 - (2ay + O(y^2)) \cdot y\,(ay^2 + O(y^3)) = 0$$

$$\underline{-ay^2}\ \underline{-O(y^3)}\ \underline{+y^2}\ \underline{-(2ay^2 + O(y^3))\,(ay^2 + O(y^3))} = 0$$

$$\underline{-ay^2}\ \underline{+y^2}\ \underline{+O(y^3)} = 0$$

$$(1-a)\,y^2 = 0$$

$$a = 1$$

$$h(y) = y^2 + O(y^3)$$

Q2: $\dot{y} = g_1(y, z)$ $\uparrow$ $z = h(y)$

$= y\,h(y)$

$= y\,(y^2 + O(y^3))$

$\therefore\ \dot{y} = y^3 + O(y^4)$ "reduced system on the manifold"

Q3: $\dot{y} = y^3 + O(y^4)$

$\Downarrow$

$x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is stable $(a = -1)$ for $\dot{x} = f(x)$

Ex.

$$\dot{x}_1 = x_2$$

$$\dot{x}_2 = -x_2 + x_1^2 + x_1 x_2$$

$$\left.\frac{\partial f}{\partial x}\right|_{x=0} = \begin{bmatrix} 0 & 1 \\ 2x_1 + x_2 & -1 + x_1 \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} = A$$

$$\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$v_1 = 1, \; v_2 = 0$$

$$A - \lambda I = \begin{bmatrix} -\lambda & 1 \\ 0 & -1-\lambda \end{bmatrix}$$

$$\det(A - \lambda I) = 0 \Rightarrow \lambda(\lambda + 1) = 0$$

$$\Rightarrow \lambda = 0, \; \lambda = -1$$

$$\begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = A \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = T \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} \qquad T \text{ is invertible}$$

$$\begin{bmatrix} \dot{y} \\ \dot{z} \end{bmatrix} = T \begin{bmatrix} \dot{x}_1 \\ \dot{x}_2 \end{bmatrix} = TA \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$$\begin{bmatrix} \dot{y} \\ \dot{z} \end{bmatrix} = TAT^{-1} \begin{bmatrix} y \\ z \end{bmatrix}$$

$$T = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}$$

$$TAT^{-1} = \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}}_{T} \underbrace{\begin{bmatrix} 0 & 1 \\ 0 & -1 \end{bmatrix}}_{A} \underbrace{\begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix}}_{T^{-1}}$$

$$= \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 \\ 0 & -1 \end{bmatrix}$$

$$\begin{bmatrix} y \\ z \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

$y = x_1 + x_2 \quad \Rightarrow x_1 = y + z$

$z = -x_2 \quad \Rightarrow x_2 = -z$

$\dot{y} = \dot{x}_1 + \dot{x}_2 = x_1^2 + x_1 x_2$

$= x_1(x_1 + x_2)$

$= (y+z) \cdot y$

$\dot{z} = -\dot{x}_2 = x_2 - x_1^2 - x_1 x_2$

$= -z - x_1 y$

$= -z - y^2 - yz$

$\dot{y} = 0 + y^2 + yz$

$\dot{z} = -z - y^2 - yz$

$\lambda_2 h(y) + g_2(y, h(y)) - \frac{\partial h}{\partial y} g_1(y, h(y)) = 0$

$(-1) h(y) + (-y^2 - yz) - \frac{\partial h}{\partial y}(y^2 + yz) = 0$

$-h(y) - \left(\frac{\partial h}{\partial y} + 1\right)\left(y^2 + y h(y)\right) = 0$

let's try $h(y) = ay^2 + O(y^3)$

$-ay^2 - O(y^3) - (2ay + O(y^2) + 1)(y^2 + ay^3 + O(y^4)) = 0$

$-ay^2 - O(y^3) = 0$

$a = 0$

$\Rightarrow h(y) = O(y^3)$

"Reduced" System: $\dot{y} = y^2 + yz$

$= y^2 + y\, h(y)$

$= y^2 + y\, O(y^3)$

$= y^2 + O(y^4)$

$y = 0$ is unstable for the reduced system

$\Rightarrow x = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ is unstable for $\dot{x} = f(x)$